Riemann's Hypothesis as an Eigenvalue Problem. II

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Dedicated to Professor E. Thoma on the occasion of his 60th birthday.

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ABSTRACT

We continue the investigations concerning the eigenvalues $\lambda_1, \ldots, \lambda_{N-1}$ of a certain matrix A_N over the integers. On one side they permit an equivalent formulation of Riemann's hypothesis, and on the other side they are fairly uniformly distributed—with some characteristic exceptions. Section 1 contains arithmetic properties of the characteristic polynomial $\chi_N(x)$ of A_N and in particular the examination of $\chi_N(x)$ at $x = 1$, which leads to a multiplicative function ρ similar to the Möbius function μ. Section 2 gives estimates for the large eigenvalues of A_N and for the power sums of all eigenvalues.

0. INTRODUCTION

The matrix

$$A_N = (a_{m,n})_{2 \leq m,n \leq N}, \qquad a_{m,n} = \begin{cases} m-1 & \text{if } m \mid n, \\ -1 & \text{if } m \nmid n, \end{cases}$$

has the determinant

$$\det A_N = N! \sum_{1 \leq n \leq N} \frac{\mu(n)}{n}, \tag{0.1}$$

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where μ denotes the Möbius function, and thus is closely connected with Riemann's hypothesis, which is true if and only if $\det A_N = O(N!N^{-1/2+\varepsilon})$ for every $\varepsilon > 0$ [4, Theorem 14.25(C); 3, Theorem 1]. Reference [3] contains the first investigations of the eigenvalues $\lambda_1, \ldots, \lambda_{N-1}$ of A_N:

$$|\lambda_n| \leqslant N - \frac{1}{N} \qquad \text{for} \quad 1 \leqslant n \leqslant N-1, \tag{0.2}$$

[3, Theorem 2], and, if the eigenvalues are indexed appropriately,

$$n \leqslant \lambda_n < n+1 \tag{0.3}$$

for $1 \leqslant n \leqslant N-1$, with at most $2\sqrt{N}$ exceptions [3, Theorem 5]. The characteristic polynomial

$$\chi_N(x) = \det(xI - A_N)$$

and its normalized form

$$\hat{\chi}_N(x) := \chi_N(x) \prod_{2 \leqslant m \leqslant N} (x-m)^{-1} \tag{0.4}$$

can be represented recursively [3, Theorem 3]:

$$\chi_N(x) = (x-N)\chi_{N-1}(x) + h_N(x), \tag{0.5}$$

$$\hat{\chi}_N(x) = \sum_{1 \leqslant n \leqslant N} \hat{h}_n(x), \tag{0.6}$$

where $h_1(x) = 1$,

$$h_N(x) = \sum_{\substack{1 \leqslant k < N \\ k \mid N}} k h_k(x) \prod_{k < m < N} (x-m) \qquad \text{for} \quad N > 1, \tag{0.7}$$

and

$$\hat{h}_N(x) = h_N(x) \prod_{2 \leqslant m \leqslant N} (x-m)^{-1}. \tag{0.8}$$

The most characteristic places of $\hat{\chi}_N(x)$ seem to be $x = 0$ and $x = 1$, for the rational functions $\hat{h}_N(x)$ are multiplicative in N exactly for these two values of x (Theorem 1). $x = 0$ leads back to the Möbius function

$$\hat{h}_n(0) = \frac{\mu(n)}{n} \quad \text{and} \quad \hat{\chi}_N(0) = \sum_{1 \leqslant n \leqslant N} \frac{\mu(n)}{n},$$

whereas $x = 1$ transforms the question of estimates for $\Sigma_{1 \leqslant n \leqslant N} \mu(n)/n$ into the problem of estimates for

$$\hat{\chi}_N(1) = \sum_{1 \leqslant n \leqslant N} \hat{h}_n(1) = \sum_{1 \leqslant n \leqslant N} \frac{\rho(n)}{n},$$

where ρ is the multiplicative function defined by

$$\rho(p^m) = (-1)^m p^m \prod_{1 \leqslant k \leqslant m} (p^k - 1)^{-1}$$

for prime powers p^m. Connected with ρ by the identity

$$\sum_{d \mid n} \rho(d) \tau\left(\frac{n}{d}\right) = \delta_{1,n}$$

(Theorem 2) is the multiplicative function τ, defined for prime powers by

$$\tau(p^m) = \prod_{1 \leqslant k \leqslant m} (1 - p^{-k})^{-1},$$

which plays a vital role in the analysis of the eigenvalues. Theorem 2 shows that ρ has the same inversion property with respect to τ as the Möbius function has with respect to the constant function 1. Therefore an analysis of the eigenvalues of A_N in terms of ρ and τ means to examine $\chi_N(x)$ at $x = 1$ instead of $x = 0$. At the first glance this shift may seem to be unsuitable, since

$$\tau(n) = O(\log_2 n), \tag{0.9}$$

which is an elementary consequence of the distribution of primes—more precisely, of $\Sigma_{p \leqslant m} 1/p = O(\log_2 m)$. But there are indications that $x = 1$ is more appropriate than $x = 0$. For instance, the values $\hat{\chi}_M(1)$ reflect the

whole polynomial $\chi_N(x)$:

$$\frac{(-1)^N}{(N-n)!}\chi_N(n) = (-1)^n n! \frac{\tau(n)}{n^2}\hat{\chi}_{[N/n]}(1), \qquad 1 \leqslant n \leqslant N, \quad (0.10)$$

[3, Theorem 4], and, as a consequence,

$$\chi_N(x) = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n}\hat{\chi}_{[N/n]}(1) \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n}} (x-m),$$

(Theorem 5). The connection between the values of $\hat{\chi}_N(x)$ at $x = 0$ and $x = 1$ is established in Theorem 4:

$$\hat{\chi}_N(1) = \sum_{1 \leqslant n \leqslant N} \frac{\rho(n)}{n^2}\hat{\chi}_{[N/n]}(0),$$

$$\hat{\chi}_N(0) = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n^2}\hat{\chi}_{[N/n]}(1).$$

This interplay can be used as the basis of conditional proofs for a (small) part of Riemann's hypothesis—which will be the contents of a subsequent paper.

Theorem 5 implies identities for the eigenvalues (Corollary 2 of Theorem 5), which, combined with estimates on sums over the τ-values (which are proved in the Appendix), are the starting point for the finer analysis of the distribution of the real eigenvalues of A_N in the intervals $[N/(m+1), N/m[$ for small m compared with N. This is done explicitly only for the case $m = 1$ (Theorem 6), which generalizes Theorem 11 in [3]: Each interval $[n-1, n[$, $[N/2]+2 \leqslant n \leqslant N$, contains exactly one eigenvalue λ_n of A_N, and they all are of order 1. In particular,

$$\lambda_n = n - \frac{\tau(n)}{\zeta_\infty \log n}\left[1 + O\left(\frac{\log_3 N}{\log_2 N}\right)\right], \qquad \text{if } n \text{ is near } N,$$

$$\lambda_n = n - 1 + \frac{\tau(n-1)}{\zeta_\infty \log n}\left[1 + O\left(\frac{\log_3 N}{\log_2 N}\right)\right], \qquad \text{if } n \text{ is near } \frac{N}{2},$$

where

$$\zeta_\infty = \sum_{n \geqslant 1} \frac{\tau(n)}{n^2} = \prod_{m \geqslant 2} \zeta(m) = 2.294856\ldots$$

(Theorem 3), and $\zeta(s)$ denotes the Riemann zeta function.

Theorem 7 deals with global estimates for all eigenvalues λ of A_N, $N \geqslant N_0$:

$$|\operatorname{Im} \lambda| < \frac{N}{3},$$

$$\operatorname{Re} \lambda > -\frac{N}{3},$$

$$\operatorname{Re} \lambda < \frac{N}{2} \qquad \text{if} \quad \operatorname{Im} \lambda \neq 0.$$

These results can be improved substantially; therefore the proofs are only sketched. Probably the maximum of the real parts of the complex eigenvalues of A_N is approximately $N/7$, as a consequence of the sign changes $\hat{\chi}_5(1) < 0$, $\hat{\chi}_6(1) > 0$, $\hat{\chi}_7(1) < 0$, which imply for $N \geqslant N_0$ that the intervals $]\,[N/6], [N/6] + 1[$and$][N/7], [N/7] + 1[$ contain no eigenvalues of A_N. In any case Theorem 7 is sufficient to fill the gap of Theorem 9 in [3]: For all $N \geqslant N_0$ and all $k \geqslant 0$,

$$\sum_{1 \leqslant n \leqslant N-1} n^k \leqslant \sum_{\lambda} \lambda^k \leqslant \sum_{2 \leqslant n \leqslant N} n^k, \tag{0.11}$$

where the λ-sum is over the $N-1$ eigenvalues of A_N. This goes well with (0.3) and indicates that the exceptional eigenvalues of A_N, i.e. those which cannot be located by the method of sign changes in Theorem 5 of [3], should obey certain rules of regular distribution. The general motion of $\Sigma \lambda^k$ as a function of k between the two power sums in (0.11) has been studied in Section 2 of [3].

1. ARITHMETIC PROPERTIES OF $\chi_N(x)$

THEOREM 1.

(1) $\hat{h}_N(x)$ *is multiplicative in* N *if and only if* $x = 0$ *or* $x = 1$.
(2) $\hat{h}_n(0) = \mu(n)/n$ *and* $\hat{\chi}_N(0) = \Sigma_{1 \leqslant n \leqslant N} \mu(n)/n$.
(3) $\hat{h}_n(1) = \rho(n)/n$ *and* $\hat{\chi}_N(1) = \Sigma_{1 \leqslant n \leqslant N} \rho(n)/n$.

Proof. Equations (0.6) and (0.1) show

$$\sum_{1 \leqslant n \leqslant N} \hat{h}_n(0) = \hat{\chi}_N(0) = \frac{1}{N!}(-1)^{N-1}\chi_N(0)$$

$$= \frac{1}{N!} \det A_N = \sum_{1 \leqslant n \leqslant N} \frac{\mu(n)}{n},$$

which proves part (2) and the multiplicativity of $\hat{h}_N(0)$. For $N > 1$ the definition of $h_N(x)$ in (0.7) gives

$$x\hat{h}_N(x) = \sum_{\substack{1 \leqslant k \leqslant N \\ k \mid N}} k\hat{h}_k(x) \tag{1.1}$$

and in particular, for $N \geqslant 1$, $\hat{h}_N(1) = \sum_{k \mid N} k\hat{h}_k(1)$. To prove $n\hat{h}_n(1) = \rho(n)$ it therefore suffices to show

$$\frac{\rho(N)}{N} = \sum_{k \mid N} \rho(k) \tag{1.2}$$

for prime powers $N = p^m$, since ρ is multiplicative. This is done by induction:

$$\sum_{0 \leqslant k \leqslant m+1} \rho(p^k) = \rho(p^m)p^{-m} + \rho(p^{m+1})$$

$$= (-1)^m \prod_{1 \leqslant k \leqslant m} (p^k - 1)^{-1}$$

$$+ (-p)^{m+1} \prod_{1 \leqslant k \leqslant m+1} (p^k - 1)^{-1}$$

$$= (-1)^{m+1} \prod_{1 \leqslant k \leqslant m+1} (p^k - 1)^{-1}$$

$$= \rho(p^{m+1})p^{-m-1}. \tag{1.3}$$

This proves part (3) and the multiplicativity of $\hat{h}_N(1)$.

Finally if $\hat{h}_N(x)$ is multiplicative in N, then in particular for prime numbers $p \neq q$

$$\hat{h}_p(x)\hat{h}_q(x) = \hat{h}_{pq}(x).$$

But Equation (1.1) shows

$$\hat{h}_p(x) = \frac{1}{x-p}, \qquad \hat{h}_{pq}(x) = \frac{1}{x-pq}\left(1 + \frac{p}{x-p} + \frac{q}{x-q}\right),$$

and this implies $x^2 = x$ and hence $x = 0$ or $x = 1$. ∎

THEOREM 2. $\Sigma_{d \mid n}\rho(d)\tau(n/d) = \delta_{1,n}$.

Proof. Since ρ and τ are multiplicative, it suffices to prove the equation for prime powers p^n. For

$$S_n := p^{-n} \sum_{0 \leqslant m \leqslant n} \tau(p^m)\rho(p^{n-m})$$

and

$$T_n := \sum_{0 \leqslant m \leqslant n} \tau(p^m)\rho(p^{n-m})p^{m-n}$$

we show

$$T_n = \sum_{0 \leqslant m \leqslant n} p^m S_m \tag{1.4}$$

and

$$T_n = \sum_{0 \leqslant m \leqslant n} S_m. \tag{1.5}$$

The difference of these equations yields by induction $S_n = 0$ for $n \geqslant 1$.

(1.4):

$$\sum_{0 \leqslant m \leqslant n} p^m S_m = \sum_{0 \leqslant m \leqslant n} \sum_{0 \leqslant r \leqslant m} \tau(p^r)\rho(p^{m-r})$$

$$= \sum_{0 \leqslant r \leqslant n} \tau(p^r)\Big(\sum_{0 \leqslant k \leqslant n-r} \rho(p^k)\Big)$$

$$= \sum_{0 \leqslant r \leqslant n} \tau(p^r)\rho(p^{n-r})p^{r-n} \qquad \text{[by (1.3)]}$$

$$= T_n.$$

(1.5): With regard to $T_0 = S_0$ it suffices to show $T_n = S_n + T_{n-1}$ for $n \geqslant 1$:

$$T_n - S_n = \sum_{0 \leqslant m \leqslant n} \left(\tau(p^m) \frac{\rho(p^{n-m})}{p^{n-m}} - \frac{\tau(p^m)}{p^m} \frac{\rho(p^{n-m})}{p^{n-m}} \right)$$

$$= \sum_{1 \leqslant m \leqslant n} \tau(p^m)(1 - p^{-m})\rho(p^{n-m})p^{m-n}$$

$$= \sum_{1 \leqslant m \leqslant n} \tau(p^{m-1})\rho(p^{n-m})p^{m-n}$$

$$= T_{n-1}. \qquad \blacksquare$$

COROLLARY.

$$\left(\sum_{n \geqslant 1} \frac{\tau(n)}{n^s} \right)\left(\sum_{n \geqslant 1} \frac{\rho(n)}{n^s} \right) = 1 \qquad \textit{for} \quad \operatorname{Re} s > 1.$$

Proof. The convergence of the series comes from the estimate $\tau(n) = O(\log_2 n)$ in (0.9), and

$$\left(\sum_{n \geqslant 1} \frac{\tau(n)}{n^s} \right)\left(\sum_{n \geqslant 1} \frac{\rho(n)}{n^s} \right) = \sum_{n \geqslant 1} \frac{1}{n^s} \left(\sum_{d \mid n} \tau\left(\frac{n}{d}\right)\rho(d) \right) = 1$$

by Theorem 2. $\blacksquare$

THEOREM 3.

(1) $\Sigma_{n \geqslant 1} \tau(n)/n^s = \prod_{m \geqslant 0} \zeta(s+m)$.
(2) $\Sigma_{n \geqslant 1} |\rho(n)|/n^s = \prod_{m \geqslant 0} \zeta(s+m)/\zeta(2(s+m))$.

Proof.

$$(1-p^{-s})\Big(\sum_{m\geqslant 0}\tau(p^m)p^{-ms}\Big)=(1-p^{-s})\Big(\sum_{m\geqslant 0}p^{-ms}\prod_{1\leqslant r\leqslant m}(1-p^{-r})^{-1}\Big)$$

$$=1+\sum_{m\geqslant 1}p^{-ms}\prod_{1\leqslant r\leqslant m}(1-p^{-r})^{-1}$$

$$-\sum_{m\geqslant 1}p^{-ms}\prod_{1\leqslant r\leqslant m-1}(1-p^{-r})^{-1}$$

$$=1+\sum_{m\geqslant 1}p^{m-ms}\prod_{1\leqslant r\leqslant m}(1-p^{-r})^{-1}$$

$$=\sum_{m\geqslant 0}\tau(p^m)p^{-m(s+1)}.$$

Multiplication of these series for all primes p gives

$$\zeta(s)^{-1}\Big(\sum_{n\geqslant 1}\frac{\tau(n)}{n^s}\Big)=\sum_{n\geqslant 1}\frac{\tau(n)}{n^{s+1}},$$

which by iteration proves part (1). Part (2) can be shown in the same way, since

$$\prod_p\Big(1+\frac{1}{p^s}\Big)=\prod_p\frac{1-p^{-2s}}{1-p^{-s}}=\frac{\zeta(s)}{\zeta(2s)},$$

and

$$(1+p^{-s})\Big(\sum_{m\geqslant 0}|\rho(p^m)|p^{-m(s+1)}\Big)=(1+p^{-s})\Big(\sum_{m\geqslant 0}p^{-ms}\prod_{1\leqslant r\leqslant m}(p^r-1)^{-1}\Big)$$

$$=\sum_{m\geqslant 0}p^{-ms}\prod_{1\leqslant r\leqslant m}(p^r-1)^{-1}$$

$$+\sum_{m\geqslant 0}p^{-(m+1)s}\prod_{1\leqslant r\leqslant m}(p^r-1)^{-1}$$

$$=1+\sum_{m\geqslant 1}p^{m-ms}\prod_{1\leqslant r\leqslant m}(p^r-1)^{-1}$$

$$=\sum_{m\geqslant 0}|\rho(p^m)|p^{-ms}.$$

■

COROLLARY.

$$\sum_{d \mid n} \frac{\rho(d)}{d} \tau\left(\frac{n}{d}\right) = 1,$$

$$\sum_{d \mid n} \rho\left(\frac{n}{d}\right) \frac{\tau(d)}{d} = \mu(n).$$

Proof. We compare the coefficients of the series

$$\sum_{n \geq 1} \frac{1}{n^s} = \zeta(s) = \left(\sum_{n \geq 1} \frac{\rho(n)}{n^{s+1}} \right) \left(\sum_{n \geq 1} \frac{\tau(n)}{n^s} \right)$$

[Corollary of Theorem 2; Theorem 3(1)]

$$= \sum_{n \geq 1} \frac{1}{n^s} \left(\sum_{d \mid n} \frac{\rho(d)}{d} \tau\left(\frac{n}{d}\right) \right)$$

and

$$\sum_{n \geq 1} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)} = \left(\sum_{n \geq 1} \frac{\rho(n)}{n^s} \right) \left(\sum_{n \geq 1} \frac{\tau(n)}{n^{s+1}} \right)$$

$$= \sum_{n \geq 1} \frac{1}{n^s} \left(\sum_{d \mid n} \rho\left(\frac{n}{d}\right) \frac{\tau(d)}{d} \right). \qquad \blacksquare$$

THEOREM 4.

$$\hat{\chi}_N(1) = \sum_{1 \leq n \leq N} \frac{\rho(n)}{n^2} \hat{\chi}_{[N/n]}(0),$$

$$\hat{\chi}_N(0) = \sum_{1 \leq n \leq N} \frac{\tau(n)}{n^2} \hat{\chi}_{[N/n]}(1).$$

Proof.

$$\hat{\chi}_N(1) = \sum_{1 \leqslant m \leqslant N} \frac{\rho(m)}{m} \qquad \text{[Theorem 1(3)]}$$

$$= \sum_{1 \leqslant m \leqslant N} \frac{1}{m} \sum_{d \mid m} \mu\left(\frac{m}{d}\right) \frac{\rho(d)}{d}$$

[since $(\sum_{n \geqslant 1} \mu(n)/n^s)(\sum_{n \geqslant 1} \rho(n)/n^{s+1}) = \sum_{n \geqslant 1} \rho(n)/n^s$ by the Corollary of Theorem 2 and Theorem 3(1)]

$$= \sum_{1 \leqslant d \leqslant N} \frac{\rho(d)}{d^2} \left(\sum_{\substack{1 \leqslant m \leqslant N \\ d \mid m}} \frac{\mu(m/d)}{m/d} \right)$$

$$= \sum_{1 \leqslant d \leqslant N} \frac{\rho(d)}{d^2} \left(\sum_{1 \leqslant n \leqslant N/d} \frac{\mu(n)}{n} \right)$$

$$= \sum_{1 \leqslant d \leqslant N} \frac{\rho(d)}{d^2} \hat{\chi}_{[N/d]}(0) \qquad \text{[Theorem 1(2)]}.$$

$$\hat{\chi}_N(0) = \sum_{1 \leqslant k \leqslant N} \hat{\chi}_{[N/k]}(0) \frac{1}{k^2} \left(\sum_{\substack{m,n \\ mn = k}} \tau(n)\rho(m) \right) \qquad \text{[Theorem 2]}$$

$$= \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n^2} \left(\sum_{1 \leqslant m \leqslant N/n} \frac{\rho(m)}{m^2} \hat{\chi}_{[N/mn]}(0) \right)$$

$$= \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n^2} \hat{\chi}_{[N/n]}(1)$$

by the first equation of the theorem. ∎

As an immediate consequence of Theorem 4 and the elementary result $|\hat{\chi}_k(0)| = |\sum_{1 \leqslant n \leqslant k} \mu(n)/n| \leqslant 1$ [2, p. 582], we have:

COROLLARY.

$$|\hat{\chi}_k(1)| \leqslant \sum_{n \geqslant 1} \frac{|\rho(n)|}{n^2} = 2.072956\ldots \qquad \textit{for all} \quad k \geqslant 1.$$

THEOREM 5.

$$(1)\qquad \chi_N(x) = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \hat{\chi}_{[N/n]}(1) \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n}} (x - m),$$

$$(2)\qquad \hat{\chi}_N(x) = (1 - x) \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n(n - x)} \hat{\chi}_{[N/n]}(1).$$

Proof. Lagrange interpolation at $n = 1, 2, 3, \ldots, N$ shows

$$\chi_N(x) = \sum_{1 \leqslant n \leqslant N} \chi_N(n) \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n}} \frac{x - m}{n - m}$$

$$= \sum_{1 \leqslant n \leqslant N} \chi_N(n) \frac{(-1)^{N-n}}{(n-1)!(N-n)!} \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n}} (x - m)$$

$$= \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \hat{\chi}_{[N/n]}(1) \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n}} (x - m) \qquad \text{[Equation (0.10)]}.$$

The second equation is a consequence of the first one and of (0.4). ∎

A comparison of the coefficients of x^{N-1} in the first equation of Theorem 5 yields at once

COROLLARY 1.

$$\sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \hat{\chi}_{[N/n]}(1) = 1.$$

This is the pendant at $x = 1$ to $\sum_{1 \leqslant n \leqslant N} (1/n) \hat{\chi}_{[N/n]}(0) = 1$.

COROLLARY 2. *For all eigenvalues* λ_1, λ_2, $\lambda = \alpha + \beta i$ *of* A_N:

$$(1)\qquad \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n - \lambda} \hat{\chi}_{[N/n]}(1) = 1$$

if $\lambda \notin \mathbb{N}$.

$$\sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{(n-\lambda_1)(n-\lambda_2)} \hat{\chi}_{[N/n]}(1) = 0 \tag{2}$$

if $\lambda_1, \lambda_2 \notin \mathbb{N}$ *and* $\lambda_1 \neq \lambda_2$.

$$\sum_{1 \leqslant n \leqslant N} \frac{\tau(n) n^{\delta}}{(n-\alpha)^2 + \beta^2} \hat{\chi}_{[N/n]}(1) = \delta \tag{3}$$

for $\delta = 0$ *and* $\delta = 1$, *if* $\beta \neq 0$.

$$\sum_{\substack{1 \leqslant n \leqslant N \\ n \neq \lambda}} \frac{\tau(n)}{n(n-\lambda)} \hat{\chi}_{[N/n]}(1) = 0 \tag{4}$$

if $\lambda \in \mathbb{N}$ *is of order* $\geqslant 2$.

$$\sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{(n-\lambda)^2} \hat{\chi}_{[N/n]}(1) = 0 \tag{5}$$

if $\lambda \notin \mathbb{N}$ *is of order* $\geqslant 2$.

Proof. (1): We substitute $x = \lambda$ in the first equation of Theorem 5, divide by $(1/\lambda)\prod_{1 \leqslant n \leqslant N}(\lambda - n)$, and find

$$0 = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \frac{\lambda}{\lambda - n} \hat{\chi}_{[N/n]}(1). \tag{1.6}$$

The difference of the equation in Corollary 1 and (1.6) is part (1).

(2) comes from part (1), since

$$\frac{1}{n-\lambda_1} - \frac{1}{n-\lambda_2} = \frac{\lambda_1 - \lambda_2}{(n-\lambda_1)(n-\lambda_2)}.$$

(3) is a consequence of part (2) with $\lambda_1 = \alpha + \beta i$, $\lambda_2 = \alpha - \beta i$, when we equate real and imaginary parts.

(4): The derivative of the first equation in Theorem 5 is

$$\chi_N'(x) = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \hat{\chi}_{[N/n]}(1) \left(\sum_{\substack{1 \leqslant t \leqslant N \\ t \neq n}} \prod_{\substack{1 \leqslant m \leqslant N \\ m \neq n, t}} (x - m) \right).$$

Here we substitute $x = \lambda$ and observe that $\chi_N'(\lambda) = 0$, since λ is a root of $\chi_N(x)$ of order $\geqslant 2$.

(5) is established by substituting $x = \lambda$ into the derivative of

$$\frac{\chi_N(x)}{\prod_{1 \leqslant n \leqslant N}(x - n)} = \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \frac{1}{x - n} \hat{\chi}_{[N/n]}(1),$$

observing that $\chi_N(\lambda) = \chi_N'(\lambda) = 0$. This yields

$$\sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n(n - \lambda)^2} \hat{\chi}_{[N/n]}(1) = 0.$$

And finally, with regard to part (1) and Corollary 1,

$$\begin{aligned} 0 &= \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n - \lambda} \hat{\chi}_{[N/n]}(1) - \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n} \hat{\chi}_{[N/n]}(1) \\ &\quad + \lambda^2 \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{n(n - \lambda)^2} \hat{\chi}_{[N/n]}(1) \\ &= \lambda \sum_{1 \leqslant n \leqslant N} \frac{\tau(n)}{(n - \lambda)^2} \hat{\chi}_{[N/n]}(1). \end{aligned}$$

■

2. ESTIMATES FOR THE EIGENVALUES

THEOREM 6. *Suppose $N \geqslant N_0$, and let $M := [N/2]$.*

(1) *M is the largest integer which is an eigenvalue of A_N.*
(2) *The interval $]M, M + 1[$ contains no eigenvalues of A_N.*

(3) *Each interval* $]n-1, n[$, $M+2 \leqslant n \leqslant N$, *contains exactly one eigenvalue* λ_n *of* A_N, *all these eigenvalues are of order* 1, *and*

$$n-1+\frac{\tau(n-1)}{3\log N} \leqslant \lambda_n \leqslant n-\frac{\tau(n)}{3\log N}.$$

(4) *In particular*:

$$\lambda_n = n-\frac{\tau(n)}{\zeta_\infty \log n}\left(1+O\left(\frac{\log_3 N}{\log_2 N}\right)\right) \quad \text{if} \quad N-\exp\left(\frac{\log N}{\log_2^2 N}\right) \leqslant n \leqslant N,$$

$$\lambda_n = n-1+\frac{\tau(n-1)}{\zeta_\infty \log(n-1)}\left(1+O\left(\frac{\log_3 N}{\log_2 N}\right)\right)$$

$$\text{if} \quad M+2 \leqslant n \leqslant M+\exp\left(\frac{\log N}{\log_2^2 N}\right).$$

Proof. (1): M is an eigenvalue of A_N by (0.10), since $\hat{\chi}_2(1)=0$; and Equation (0.2) asserts that all eigenvalues are $< N$. But the integers n with $M < n < N$ cannot be eigenvalues by (0.10), since $\hat{\chi}_1(1)=1 \neq 0$.

(2): Suppose λ is an eigenvalue of A_N and $M < \lambda < M+1$. Then Corollary 2(1) of Theorem 5, combined with $\hat{\chi}_1(1)=1$ and $\hat{\chi}_2(1)=0$, shows

$$\sum_{N/2<n\leqslant N} \frac{\tau(n)}{n-\lambda} = 1+\sum_{1\leqslant n\leqslant N/3} \frac{\tau(n)}{\lambda-n}\hat{\chi}_{[N/n]}(1).$$

Here we have on one side

$$\sum_{1\leqslant n\leqslant N/3} \frac{\tau(n)}{\lambda-n}\hat{\chi}_{[N/n]}(1) \leqslant \frac{3}{N/6}\sum_{1\leqslant n\leqslant N/3}\tau(n) \leqslant 18,$$

by the Corollary of Theorem 4, the τ-estimate in the Corollary of the Appendix, and the fact that $\lambda > M$. But on the other side,

$$\sum_{N/2<n\leqslant N} \frac{\tau(n)}{n-\lambda} \geqslant \sum_{1\leqslant n<N/2} \frac{1}{n},$$

which is a contradiction if $N \geqslant N_0$.

(3): Theorem 5(2) in [3] asserts that each interval $]n-1, n[$, $M+2 \leqslant n \leqslant N$, contains at least one eigenvalue. Suppose now that there are two eigenvalues $\lambda_1 \neq \lambda_2$ in $]n-1, n[$ or one eigenvalue $\lambda_1 = \lambda_2$ which is at least twofold. In both cases Corollary 2 of Theorem 5 gives

$$\sum_{1 \leqslant m \leqslant N} \frac{\tau(m)}{(m-\lambda_1)(m-\lambda_2)} \hat{\chi}_{[N/m]}(1) = 0,$$

that is, as before,

$$\sum_{N/2 < m \leqslant N} \frac{\tau(m)}{(m-\lambda_1)(m-\lambda_2)} = -\sum_{1 \leqslant m \leqslant N/3} \frac{\tau(m)}{(m-\lambda_1)(m-\lambda_2)} \hat{\chi}_{[N/m]}(1),$$

which again is impossible if $N \geqslant N_0$, since the left hand side is > 1, but the right hand sum is $O(1/N)$.

Now let $\lambda_n = n - \varepsilon_n$ denote the eigenvalue in $]n-1, n[$. Then by Corollary 2(1) of Theorem 5,

$$\sum_{N/2 < m \leqslant N} \frac{\tau(m)}{m-\lambda_n} = O(1),$$

again because the values $\hat{\chi}_k(1)$ are bounded and $\hat{\chi}_1(1) = 1$, $\hat{\chi}_2(1) = 0$. A shift of indices and use of the estimate $\tau(n) = O(\log_2 n)$ gives

$$\frac{\tau(n)}{\varepsilon_n} - \frac{\tau(n-1)}{1-\varepsilon_n} = \sum_{2 \leqslant m < n-M} \frac{\tau(n-m)}{m} - \sum_{1 \leqslant m \leqslant N-n} \frac{\tau(n+m)}{m} + O(\log_2 N). \qquad (2.1)$$

If we now assume $\varepsilon_n \leqslant \frac{1}{2}$, we find

$$\frac{\tau(n)}{\varepsilon_n} \leqslant \sum_{2 \leqslant m < n} \frac{\tau(n-m)}{m} + c_1 \log_2 N$$

$$\leqslant 3 \log N \qquad \text{by the Corollary in the Appendix.}$$

And if $\varepsilon_n > \frac{1}{2}$, then (2.1) implies

$$\frac{\tau(n-1)}{1-\varepsilon_n} \leqslant \sum_{1 \leqslant m \leqslant N} \frac{\tau(n+m)}{m} + c_2 \log_2 N$$

$$\leqslant 3 \log N,$$

again by the Corollary in the Appendix.

(4) as well is a consequence of Equation (2.1):

$$N - \exp\left(\frac{\log N}{\log_2^2 N}\right) \leqslant n \leqslant N \quad \text{and} \quad \tau(m) = O(\log_2 m)$$

imply

$$\sum_{1 \leqslant m \leqslant N-n} \frac{\tau(n+m)}{m} = O\left(\frac{\log N}{\log_2 N}\right),$$

and then (2.1) and the Corollary in the Appendix show

$$\frac{\tau(n)}{\varepsilon_n} - \frac{\tau(n-1)}{1-\varepsilon_n} = \sum_{2 \leqslant m < n-M} \frac{\tau(n-m)}{m} + O\left(\frac{\log N}{\log_2 N}\right)$$

$$= \zeta_\infty(\log(n-M))\left(1 + O\left(\frac{\log_3 N}{\log_2 N}\right)\right)$$

$$= \zeta_\infty(\log n)\left(1 + O\left(\frac{\log_3 N}{\log_2 N}\right)\right).$$

Therefore $\varepsilon_n < \frac{1}{2}$, and hence

$$\frac{\tau(n)}{\varepsilon_n} = \zeta_\infty(\log n)\left(1 + O\left(\frac{\log_3 N}{\log_2 N}\right)\right).$$

Finally, if

$$M + 2 \leqslant n \leqslant M + \exp\left(\frac{\log N}{\log_2^2 N}\right),$$

then similarly

$$\frac{\tau(n)}{\varepsilon_n} - \frac{\tau(n-1)}{1-\varepsilon_n} = -\sum_{1 \leqslant m \leqslant N-n} \frac{\tau(n+m)}{m} + O\left(\frac{\log N}{\log_2 N}\right),$$

which shows at once $\varepsilon_n > \frac{1}{2}$, and then, with the τ-estimate of the Appendix,

$$\frac{\tau(n-1)}{1-\varepsilon_n} = \zeta_\infty(\log(N-n))\left(1+O\left(\frac{\log_3 N}{\log_2 N}\right)\right)$$

$$= \zeta_\infty(\log(n-1))\left(1+O\left(\frac{\log_3 N}{\log_2 N}\right)\right). \quad \blacksquare$$

Riemann's hypothesis $|\prod_{2 \leqslant n \leqslant N} \lambda_n| = |\det A_N| = O(N! N^{-1/2+\varepsilon})$ asserts that on average

$$\lambda_n \sim n - \tfrac{1}{2}, \qquad 2 \leqslant n \leqslant N.$$

Theorem 6 suggests to consider just the eigenvalues $\lambda_n > M$ of A_N, i.e. $\lambda_n \in]n-1, n[$, $M+2 \leqslant n \leqslant N$. We compare them all with the values $n - \frac{1}{2}$ by forming the quotient

$$q_N := \prod_{M+2 \leqslant n \leqslant N} \frac{\lambda_n}{n-\frac{1}{2}}.$$

q_N can oscillate between

$$\prod_{M+2 \leqslant n \leqslant N} \frac{n-1}{n-\frac{1}{2}} \sim \frac{1}{\sqrt{2}} = 0.707\ldots \quad \text{and}$$

$$\prod_{M+2 \leqslant n \leqslant N} \frac{n}{n-\frac{1}{2}} \sim \sqrt{2} = 1.414\ldots.$$

The resulting q_N-values in Table 1 indicate that at least the eigenvalues in $]N/2, N[$—which are one-half of all eigenvalues of A_N—contribute their fair part to the truth of the Riemann hypothesis.

TABLE 1

N	q_N	N	q_N	N	q_N
10	1.018	100	0.994	190	1.000
20	0.992	110	0.999	200	0.994
30	1.000	120	0.990	210	0.994
40	0.991	130	0.998	220	0.991
50	0.996	140	0.996	230	0.998
60	0.989	150	0.997	240	0.992
70	1.004	160	0.994	250	0.999
80	0.992	170	0.996	260	0.996
90	0.995	180	0.993	270	0.996

THEOREM 7. *For all eigenvalues* λ *of* A_N, $N \geq N_0$:

(1) $|\operatorname{Im}\lambda| < N/3$,
(2) $\operatorname{Re}\lambda > -N/3$,
(3) $\operatorname{Re}\lambda < N/2$ *if* $\operatorname{Im}\lambda \neq 0$.

Proof. (1): We suppose that $\lambda = \alpha + \beta i$ is a complex eigenvalue of A_N, and $\beta_0 := |\beta|/N$. Corollary 2(3) of Theorem 5 yields

$$\sum_{1 \leq k \leq N} \hat{\chi}_k(1) \sum_{N/(k+1) < n \leq N/k} \frac{\tau(n)(N-n)}{(\alpha-n)^2+\beta^2} = -1. \tag{2.2}$$

We split the sum over k into the three sums S_+ [over $1 \leq k \leq k_0$ such that $\hat{\chi}_k(1) > 0$], S_- [over $1 \leq k \leq k_0$ such that $\hat{\chi}_k(1) < 0$], and R [over $k_0 < k \leq N$], where k_0 is a constant. Then we choose $c_0 > 0$ such that

$$|\alpha - n| \leq c_0 N \qquad \text{for} \quad 1 \leq n \leq N.$$

The τ-estimates in the Appendix now show that for all $k \leq k_0$ and $N \geq N_0$,

$$\sum_{N/(k+1) < n \leq N/k} \frac{\tau(n)(N-n)}{(\alpha-n)^2+\beta^2} \begin{cases} \geq \dfrac{\zeta_\infty}{c_0^2+\beta_0^2} \dfrac{2k^2-1}{2k^2(k+1)^2}\left(1 - \dfrac{c_1}{\log_2 N}\right), \\ \leq \dfrac{\zeta_\infty}{\beta_0^2} \dfrac{2k^2-1}{2k^2(k+1)^2}\left(1 + \dfrac{c_2}{\log_2 N}\right), \end{cases}$$

and, with the knowledge of the values $\hat{\chi}_k(1)$, $k \leqslant k_0$, we find

$$S_+ \geqslant \frac{\zeta_\infty}{c_0^2+\beta_0^2}\left(1-\frac{c_1}{\log_2 N}\right)S'_+,$$

$$S'_+ = \sum_{\substack{k\leqslant k_0\\ \hat{\chi}_k(1)>0}} \hat{\chi}_k(1)\frac{2k^2-1}{2k^2(k+1)^2} = 0.131043\ldots;$$

$$S_- \geqslant \frac{\zeta_\infty}{\beta_0^2}\left(1+\frac{c_2}{\log_2 N}\right)S'_-,$$

$$S'_- = \sum_{\substack{k\leqslant k_0\\ \hat{\chi}_k(1)<0}} \hat{\chi}_k(1)\frac{2k^2-1}{2k^2(k+1)^2} = -0.053670\ldots;$$

$$R \geqslant -\frac{\delta_0}{\beta_0^2}\zeta_\infty\left(\frac{1}{k_0}-\frac{1}{2k_0^2}\right)\left(1+\frac{c_3}{\log_2 N}\right),$$

$$\delta_0 = \max_{k>k_0}|\hat{\chi}_k(1)| < 2.073 \qquad \text{[Corollary of Theorem 4]}.$$

This is substituted in Equation (2.2) and then gives

$$\frac{\zeta_\infty}{c_0^2+\beta_0^2}S'_+ + \frac{\zeta_\infty}{\beta_0^2}S'_- - \frac{\zeta_\infty}{\beta_0^2}R' + 1 \leqslant 0,$$

where $R' = \delta_0/k_0$, and hence

$$\beta_0^2 \leqslant -A/2+\sqrt{A^2/4+B} \tag{2.3}$$

with

$$A = \zeta_\infty(S'_+ + S'_- - R') + c_0^2, \qquad B = c_0^2\zeta_\infty(R' - S'_-).$$

Equation (0.2) shows that we can choose $c_0 = 2$, and then (2.3) gives $|\beta_0| < 0.339$ when k_0 is large and accordingly R' small enough. This preliminary result will be used to prove part (2) of the theorem. Afterwards we can choose $c_0 = \frac{4}{3}$ here, and this finally leads to $|\beta_0| < \frac{1}{3}$.

(2): Here we start with the equation

$$\sum_{1 \leqslant k \leqslant N} \hat{\chi}_k(1) \sum_{N/(k+1) < n \leqslant N/k} \frac{\tau(n)(n-\alpha)}{(n-\alpha)^2+\beta^2} = 1 \tag{2.4}$$

for an eigenvalue $\lambda = \alpha + \beta i$ of A_N such that $\alpha < 0$, which is Corollary 2(3) of Theorem 5 if $\beta \neq 0$ and Corollary 2(1) if $\beta = 0$. As in part (1), we split the sum over k into the three sums S_+, S_-, and R. These are estimated with the τ-lemma in the Appendix, and moreover to estimate S_- we use the preliminary result $|\beta| < 0.339N$ of part (1). If then $\alpha \leqslant -N/3$, Equation (2.4) implies a contradiction.

(3): As in part (1), we start with Equation (2.2), suppose $\alpha \geqslant N/2$ for the complex eigenvalue $\lambda = \alpha + \beta i$ of A_N, and fix k_0. Then, since $\hat{\chi}_1(1) = 1$,

$$\sum_{N/2 < n \leqslant N} \frac{\tau(n)(N-n)}{(\alpha-n)^2+\beta^2}$$

$$- \sum_{\substack{2 \leqslant k \leqslant N \\ \hat{\chi}_k(1) < 0}} |\hat{\chi}_k(1)| \sum_{N/(k+1) < n \leqslant N/k} \frac{\tau(n)(N-n)}{(\alpha-n)^2} < -1.$$

The partial sum over $k_0 \leqslant k \leqslant N$ of the double sum is $O(1/k_0)$. To estimate the first sum we use, from part (1), $|\beta| < N/3$, and again we call upon the τ-lemma in the Appendix. Then $\alpha \geqslant N/2$ leads to a contradiction. ∎

THEOREM 8. *For all $N \geqslant N_0$ and all $k \geqslant 0$,*

$$\sum_{1 \leqslant n \leqslant N-1} n^k \leqslant \sum_{\lambda} \lambda^k \leqslant \sum_{2 \leqslant n \leqslant N} n^k,$$

where the λ-sum is over the $N-1$ eigenvalues of A_N.

Proof. In view of [3, Theorem 9], it only remains to show

$$\sum_{2 \leqslant n \leqslant N} n^k - \sum_{\lambda} \lambda^k \geqslant 0 \qquad \text{for} \quad k \geqslant \sqrt{N}.$$

By [3, Theorem 5(3)] there are at least $N - 2\sqrt{N}$ intervals $[n, n+1[$, $1 \leqslant n \leqslant N-1$, which contain an eigenvalue λ_n of A_N. For all these the differences

$(n+1)^k - \lambda_n^k$ are positive, and in particular, by Theorem 6(4),

$$N^k - \lambda_{N-1}^k \geqslant N^k - \left(N - \frac{1}{3\log N}\right)^k.$$

Now we have not yet taken into consideration at most $2\sqrt{N}$ eigenvalues, but for all those λ, we have by Theorem 7

$$|\lambda| < N\sqrt{\tfrac{1}{4}+\tfrac{1}{9}} < \tfrac{2}{3}N,$$

since by Theorem 6 all eigenvalues in $][N/2], N[$ are of order 1 and therefore have been considered before. Hence it suffices to prove

$$N^k - \left(N - \frac{1}{3\log N}\right)^k \geqslant 2\sqrt{N}\left(\tfrac{2}{3}N\right)^k,$$

or

$$1 - \left(1 - \frac{1}{3N\log N}\right)^k \geqslant 2\sqrt{N}\left(\tfrac{2}{3}\right)^k. \tag{2.5}$$

If $k \geqslant 3N\log N$, then

$$\left(1 - \frac{1}{3N\log N}\right)^k \leqslant \left(1 - \frac{1}{3N\log N}\right)^{3N\log N} \leqslant \frac{1}{e},$$

and this implies (2.5). If $k < 3N\log N$, then the mean value theorem shows

$$\begin{aligned} 1 - \left(1 - \frac{1}{3N\log N}\right)^k &= \frac{k}{3N\log N}\left(1 - \frac{\Theta}{3N\log N}\right)^{k-1} \qquad (0 \leqslant \Theta \leqslant 1) \\ &\geqslant \frac{k}{3N\log N}\left(1 - \frac{1}{3N\log N}\right)^{3N\log N} \\ &\geqslant \frac{k}{12N\log N}, \end{aligned}$$

and it remains to prove

$$\frac{k}{12N\log N} \geqslant 2\sqrt{N}\left(\tfrac{2}{3}\right)^k,$$

which is obviously true for $k \geqslant \sqrt{N} \geqslant \sqrt{N_0}$. ∎

APPENDIX

LEMMA.

$$(1) \qquad \sum_{1 \leqslant m \leqslant M} \tau(K+m) \geqslant \zeta_\infty M\left(1 - \frac{c_1}{\log M}\right),$$

$$(2) \qquad \sum_{1 \leqslant m \leqslant M} \tau(K+m) \leqslant \zeta_\infty M \frac{\log_2(K+M)}{\log_2 M}\left(1 + \frac{c_2}{\log_2 M}\right),$$

for all natural numbers K and suitable positive constants c_1 and c_2.

Proof. We need two elementary results on the distribution of prime numbers: a formula of Mertens,

$$\sum_{p \leqslant x} \frac{1}{p} = \log_2 x + C + O\left(\frac{1}{\log x}\right) \tag{A.1}$$

[1, p. 81], and a weak version of Chebyshev's theorem,

$$\frac{x}{2} \leqslant \sum_{p \leqslant x} \log p \leqslant \sum_{1 \leqslant n \leqslant x} \Lambda(n) \leqslant 3x \tag{A.2}$$

[1, p. 67], where Λ denotes the von Mangoldt function

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^m \text{ is a prime power,} \\ 0 & \text{otherwise.} \end{cases}$$

(1): We start with the module

$$P := P(M) = \exp\left(\sum_{1 \leqslant n \leqslant \frac{1}{4}\log M} \Lambda(n)\right) =: \prod_{p \mid P} p^{a_p}.$$

Chebyshev's theorem asserts

$$P \leqslant M^{3/4}.$$

Let q be a divisor of P. Then exactly $\phi(P/q)$ of the numbers $1,2,3,\ldots,P$ are divisible by q but by no q' such that $q' > q$ and $q \mid q' \mid P$, where ϕ is the Euler function. Therefore we have

$$\begin{aligned}\sum_{1 \leqslant m \leqslant M} \tau(K+m) &\geqslant \sum_{q \mid P} \phi\left(\frac{P}{q}\right)\left[\frac{M}{P}\right]\tau(q) \\ &\geqslant (M-P)\sum_{q \mid P}\frac{1}{P}\phi\left(\frac{P}{q}\right)\tau(q) \\ &= (M-P)\prod_{p \mid P} T_p,\end{aligned}$$

with

$$\begin{aligned}T_p &= p^{-a_p}\sum_{0 \leqslant b \leqslant a_p}\tau(p^b)\phi(p^{a_p-b}) \\ &= \left(1-\frac{1}{p}\right)\left(\sum_{0 \leqslant b < a_p}\tau(p^b)p^{-b} + \left(1-\frac{1}{p}\right)^{-1}\tau(p^{a_p})p^{-a_p}\right),\end{aligned}$$

[since $\phi(p^m) = p^m(1-1/p)$ if $m > 0$, and $\phi(1) = 1$]

$$\geqslant \left(1-\frac{1}{p}\right)\left(\sum_{b \geqslant 0}\tau(p^b)p^{-b} - (\tau(p^\infty) - \tau(p^{a_p}))\sum_{b > a_p} p^{-b}\right),$$

since $\tau(p^n)$ increases with n and converges to $\prod_{m \geqslant 1}(1-p^{-m})^{-1} =: \tau(p^\infty)$. Further,

$$(\tau(p^\infty) - \tau(p^{a_p}))\sum_{b > a_p} p^{-b} = O\left(p^{-2(a_p+1)}\right),$$

and, as shown in the proof of Theorem 3(1),

$$\left(1-\frac{1}{p}\right)\sum_{b \geqslant 0}\tau(p^b)p^{-b} = \sum_{b \geqslant 0}\tau(p^b)p^{-2b} = \prod_{m \geqslant 2}(1-p^{-m})^{-1} =: \zeta_\infty^{(p)},$$

which is the p-part of $\zeta_\infty = \prod_p \zeta_\infty^{(p)}$. Altogether we get

$$\sum_{1 \leqslant m \leqslant M} \tau(K+m) \geqslant (M-P)\Big(\prod_{p \mid P} \zeta_\infty^{(p)}\Big)\Big(\prod_{p \mid P}\big(1 - c_3 p^{-2(a_p+1)}\big)\Big)$$

$$\geqslant (M - M^{3/4})\Big(\zeta_\infty - \frac{c_4}{\log M}\Big)$$

$$\times\Big(1 - c_5 \sum_{p \mid P} p^{-2(a_p+1)}\Big), \tag{A.3}$$

since

$$1 \leqslant \prod_{p \nmid P} \zeta_\infty^{(p)} \leqslant \prod_{p > \frac{1}{4}\log M} \prod_{m \geqslant 2} (1 - p^{-m})^{-1} \leqslant \prod_{p > \frac{1}{4}\log M} \exp\Big(\frac{c_6}{p^2}\Big)$$

$$\leqslant \exp\Big(c_7 \sum_{p > \frac{1}{4}\log M} p^{-2}\Big) \leqslant \exp\Big(\frac{c_8}{\log M}\Big) \leqslant 1 + \frac{c_9}{\log M}.$$

Finally, the definition of P shows $p^{a_p+1} > \frac{1}{4}\log M$, which implies

$$\sum_{p \mid P} p^{-2(a_p+1)} = O\Big(\frac{1}{\log M}\Big),$$

and this, combined with (A.3), proves part (1).

(2): We start with the same module $P = P(M)$. Let $M(q)$ denote the set of numbers t, $K+1 \leqslant t \leqslant K+M$, which are divisible by q but by no q' such that $q' > q$ and $q \mid q' \mid P$. Then

$$\sum_{1 \leqslant m \leqslant M} \tau(K+m) = \sum_{q \mid P} \sum_{t \in M(q)} \tau(t). \tag{A.4}$$

We factorize $t \in M(q)$: $t = q\hat{q}r$, where $\hat{q}$ contains only prime divisors of P and $(r, P) = 1$, and receive the estimate

$$\tau(t) \leqslant \tau(q)\hat{C}\tau(r), \tag{A.5}$$

with

$$\hat{C} = \prod_{p \mid P} \prod_{m > a_p} (1 - p^{-m})^{-1} \leqslant \prod_{p \mid P} \left(1 - c_{10} p^{-a_p - 1}\right)^{-1}$$

$$\leqslant \prod_{p \mid P} \left(1 + \frac{c_{11}}{\log M}\right) \qquad \left[\text{since} \quad p^{a_p + 1} > \tfrac{1}{4} \log M\right]$$

$$\leqslant 1 + \frac{c_{12}}{\log_2 M}, \tag{A.6}$$

since the number of different prime divisors of P is $O((\log M)/(\log_2 M))$ by Chebyshev's theorem.

To estimate the third factor in (A.5) we observe that $\tau(r)$ with $r \leqslant K + M$ and $(r, P) = 1$ reaches its maximum if r is a product of as many different primes as possible, i.e. if

$$r = \prod_{\frac{1}{4} \log M < p \leqslant T} p$$

with the greatest T such that $r \leqslant K + M$. This implies

$$\sum_{\frac{1}{4} \log M < p \leqslant T} \log p \leqslant \log(K + M),$$

and, by Chebyshev's theorem,

$$T \leqslant c_{13} \log(K + M). \tag{A.7}$$

This shows

$$\tau(r) \leqslant \prod_{\frac{1}{4} \log M < p \leqslant T} \left(1 - \frac{1}{p}\right)^{-1} \leqslant \prod_{\frac{1}{4} \log M < p \leqslant T} \left(1 + \frac{1}{p} + \frac{2}{p^2}\right)$$

$$\leqslant \exp\left(2 \sum_p \frac{1}{p^2}\right) \exp\left(\sum_p \frac{1}{p}\right),$$

and

$$\exp\left(2\sum_{\frac{1}{4}\log M<p}\frac{1}{p^2}\right)\leqslant\exp\left(\frac{c_{14}}{\log M}\right)\leqslant 1+\frac{c_{15}}{\log M},$$

$$\exp\left(\sum_{\frac{1}{4}\log M<p\leqslant T}\frac{1}{p}\right)\leqslant\exp\Big\{\log_2(c_{13}\log(K+M))$$

$$-\log_2\left(\tfrac{1}{4}\log M\right)+\frac{c_{16}}{\log_2 M}\Big\}$$

[by Mertens's formula (A.1) and (A.7)]

$$\leqslant\left(1+\frac{c_{17}}{\log_2 M}\right)\frac{\log_2(K+M)}{\log_2 M}.$$

Combining, we have

$$\tau(r)\leqslant\left(1+\frac{c_{18}}{\log_2 M}\right)\frac{\log_2(K+M)}{\log_2 M}. \tag{A.8}$$

The estimates (A.6) and (A.8) are substituted in (A.5), and then Equation (A.4) gives

$$\sum_{1\leqslant m\leqslant M}\tau(K+m)\leqslant\left(1+\frac{c_{19}}{\log_2 M}\right)\frac{\log_2(K+M)}{\log_2 M}\sum_{q\mid P}\sum_{t\in M(q)}\tau(q).$$

Finally we treat the sum on the right side in a similar way as in the proof of part (1):

$$\sum_{q\mid P}\sum_{t\in M(q)}\tau(q)\leqslant(M+P)\sum_{q\mid P}\phi\left(\frac{P}{q}\right)\frac{1}{P}\tau(q)$$

$$\leqslant(M+P)\prod_{p\mid P}\left(1-\frac{1}{p}\right)\left(\sum_{0\leqslant b<a_p}\tau(p^b)p^{-b}+\left(1-\frac{1}{p}\right)^{-1}\tau(p^{a_p})p^{-a_p}\right)$$

$$\leqslant(M+P)\prod_{p\mid P}\left(1-\frac{1}{p}\right)\left(\sum_{b\geqslant 0}\tau(p^b)p^{-b}\right)$$

$$\leqslant(M+P)\prod_{p\mid P}\zeta_\infty^{(p)}$$

$$\leqslant (M + M^{3/4})\zeta_\infty. \qquad \blacksquare$$

COROLLARY. *Let $C > 1$ be any large constant.*

$$(1) \quad \sum_{1 \leqslant m \leqslant M} \tau(K+m) = \zeta_\infty M\left(1 + O\left(\frac{1}{\log_2 M}\right)\right) \qquad \text{if} \quad K \leqslant M^C.$$

$$(2) \quad \sum_{1 \leqslant m \leqslant M} \frac{\tau(K+m)}{m} = \zeta_\infty(\log M)\left(1 + O\left(\frac{\log_3 M}{\log_2 M}\right)\right) \qquad \text{if} \quad K \leqslant CM.$$

$$(3) \quad \sum_{1 \leqslant m \leqslant M} \frac{\tau(K-m)}{m} = \zeta_\infty(\log M)\left(1 + O\left(\frac{\log_3 M}{\log_2 M}\right)\right)$$

$$\text{if} \quad M < K \leqslant CM.$$

Proof. (1) is an immediate consequence of the two estimates in the Lemma. (2) and (3) can be obtained by partial summation:

$$\sum_{1 \leqslant m \leqslant M} \frac{\tau(K \pm m)}{m} = \frac{1}{M} \sum_{1 \leqslant m \leqslant M} \tau(K \pm m)$$

$$+ \sum_{1 \leqslant N < M} \left(\sum_{1 \leqslant m \leqslant N} \tau(K \pm m) \right) \frac{1}{N(N+1)}.$$

Here the short inner sums with $N \leqslant \exp((\log M)/(\log_2^2 M))$ are estimated with $\tau(n) = O(\log_2 n)$ and contribute

$$\sum_{N \leqslant \exp((\log M)/(\log_2^2 M))} \frac{c \log_2 M}{N+1} = O\left(\frac{\log M}{\log_2 M}\right)$$

to the sum on the left side. The long sums with $N > \exp((\log M)/(\log_2^2 M))$ can be dealt with by means of the Lemma. $\blacksquare$

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